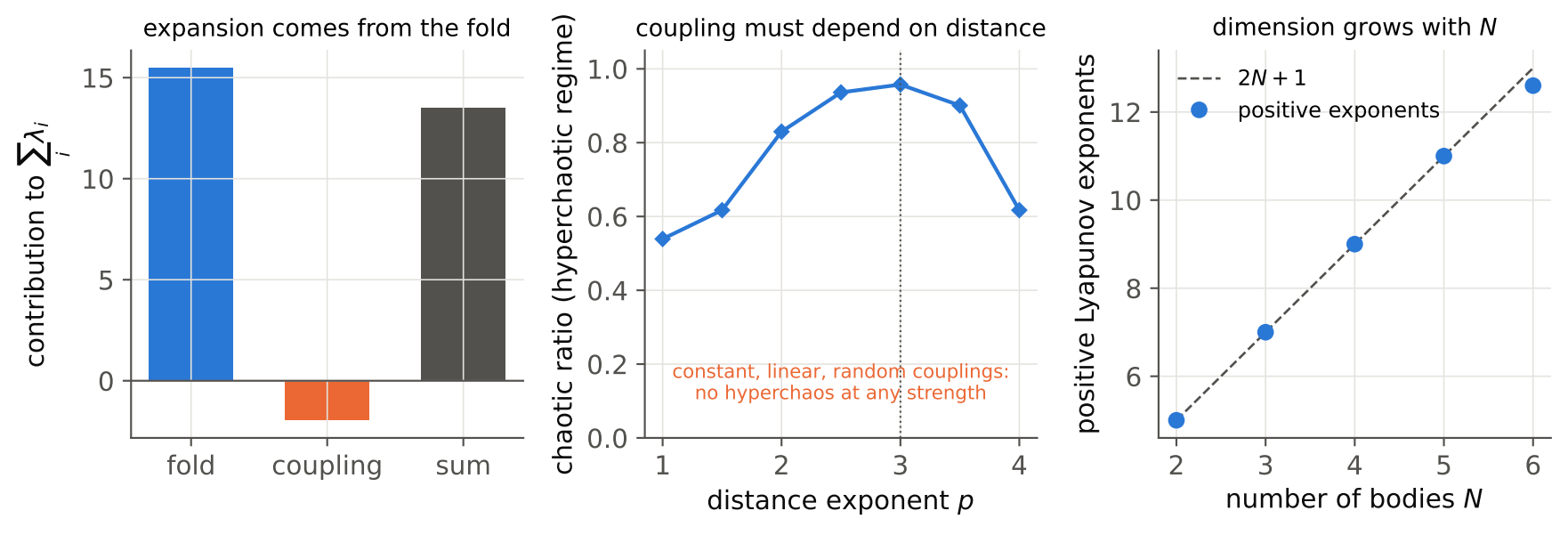
expansion comes from the fold
contribution to $\sum_i \lambda_i$
15
10
5
0
fold
coupling
sum
coupling must depend on distance
chaotic ratio (hyperchaotic regime)
1.0
0.8
0.6
0.4
0.2
0.0
constant, linear, random couplings:
no hyperchaos at any strength
1
2
3
4
distance exponent p
dimension grows with N
positive Lyapunov exponents
$2N+1$
positive exponents
12
10
8
6
2
3
4
5
6
number of bodies N